Escape speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain escape in an ideal gravitational-energy model.

SUCCESS CRITERIA - I can...

1. State the zero-at-infinity energy convention.
2. Distinguish escape speed from circular-orbit speed.
3. Name neglected effects in the ideal calculation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

Total kinetic energy changes from 15 J to 12 J in an isolated collision. Is it elastic? Where could the difference go?

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $v_{escape} = \sqrt{(2\ GM/r)}$; $v_{circular} = \sqrt{(GM/r)}$. Ideal spherical body; ignore atmosphere/rotation. Minimum escape reaches infinity at zero speed with no further thrust.

Worked reasoning: For a toy planet with $GM=50\ m^3/s^2$ and radius 4 m, escape speed is $\sqrt{(2\ GM/R)}=5$ m/s. At that limiting speed the ideal final speed at infinity is zero.

Your turn: Draw initial kinetic and gravitational-potential-energy bars whose total is zero.

Marzano check - circle: ① ② ③

03 Look and explain

Does escaping mean that gravity turns off at the launch point? Explain what the speed allows in an energy model.

Marzano check - circle: ① ② ③

04 Build the solution

A model needs 50 J to reach infinity. A projectile begins with 60 J kinetic energy. What energy is left at infinity, neglecting losses?

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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05 Use the idea

Why is an ideal escape-speed calculation not a complete rocket-launch plan?

Marzano check - circle: 1 2 3

06 Choose a method

Lift a 6 kg object by 8 m. Use g=10 N/kg. Find the change in gravitational potential energy.

Marzano check - circle: 1 2 3

07 Exit check

An object has less energy than required to escape and receives no further thrust. Can this ideal model expect it to escape?

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

The same toy planet's circular orbital speed at its surface is supplied as 3.54 m/s. Compare it with escape speed 5 m/s and explain their different purposes.

Marzano check - circle: 1 2 3